METRIC

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MILITARY SPECIFICATION

CARTRIDGE, 25MM, TARGET PRACTICE DISCARDING SABOT WITH TRACER (TPDS-T), M910

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for fabrication and packing for the Cartridge, 25mm, Target Practice Discarding Sabot with Tracer (TPDS-T), M910 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-50603	-	Primer, Percussion, M115
MIL-P-71136	-	Process Specification, Ammunition, 25mm, Packaging, Packing and Marking

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-651	-	Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US ARMY TEST AND EVALUATION COMMAND (TECOM)

ITOP 4-2-602	-	Rough Handling Tests
ITOP 4-2-805	-	Projectile Velocity and Time of Flight Measurements
MTP 4-2-827	-	Time of Flight and Ballistic Coefficients

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DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9391399	-	Interface Control Drawing, Max. Profile and Alignment, Cartridge, M791
12013671	-	Shipping Container, Linked, 25mm Ammunition (50 rounds)
12013697	-	Interface Control Drawing, 25mm Weapon/Ammunition
12524001	-	Barrel Assembly
12524003	-	Prime Item Product Fabrication Specification for Gun, Automatic, 25mm M242 with Equipment
12556500	-	Cartridge, 25mm, TPDS-T, M910
12929427	-	Packing and Marking for Container, Ammo, PA125, for Linked 25mm Ammunition

PUBLICATIONS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

AS12013566	-	25 Millimeter Ammunition Ballistic Test Methods

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3 and 6.2).

3.2 Material, design and construction. The cartridge material, design and construction shall be in accordance with MIL-A-2550 and Drawing (dwg.) 12556500. The primer used in the cartridge shall be the M115 Percussion Primer conforming to MIL-P-50603.

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3.2.1 Manufacturing process. All parts and assemblies shall be manufactured by a conventional manufacturing process. Any deviations from conventional manufacturing processes shall be approved by the contracting officer. Any change in source of supply shall be approved by the contracting officer.

3.3 Weapon/ammunition interface.

3.3.1 25mm M242 automatic weapon. The cartridge shall have compatible interface with the 25mm M242 Automatic weapon, shall not degrade the M242 reliability requirements as specified in 12524003, and shall not contribute to potential safety hazards.

3.3.2 25mm weapon ammunition. Performance and acceptance as required in this specification apply only when tests are conducted in weapons which conform to the requirements of dwg. 12013697, and autogun test barrels, in accordance with dwg. 12524001. Testing from a firm fixed mount is required.

3.4 Projectile torque. The projectile shall be capable of withstanding a minimum torque of 30 inch-pounds without rotational movement with respect to the cartridge case (see 4.5.2).

3.5 Projectile extraction. The tensile force necessary to extract the projectile from the cartridge case shall be as specified on Drawing 12556500 (see 4.5.3).

3.6 Propellant contamination. The propellant shall be free of all foreign matter (see 4.5.4).

3.7 Propellant weight. The loaded cartridge propellant weight shall be such that the requirements of 3.8, 3.9 and 3.10 are satisfied.

3.8 Chamber pressure. The average plus 3.0 standard deviations of peak chamber pressure shall not exceed 454 megapascals (MPa) when the cartridge is conditioned and then fired at 21° ± 3°C from a Mann Barrel. The average plus 2.0 standard deviations shall not exceed 496 MPa, with no individual measurement exceeding 496 MPa, when the cartridge is conditioned at any other temperature in the range from -54°C to 71°C for First Article Inspection and in the range of -32°C to +52°C for Quality Conformance Inspection. Pressure shall be measured with a piezoelectric type pressure transducer (see 4.5.5).

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3.9 Action time. The action time of the cartridge shall not exceed 6.0 milliseconds when conditioned at any temperature from -54°C to 71°C and fired from a Mann Barrel during First Article Inspection and at any temperature from -32°C to +52°C and fired from a Mann barrel during Quality Conformance Inspection. Action time is defined as the time period between the initial contact of the weapon firing pin against the primer and the exit of the projectile from the muzzle (see 4.5.6).

3.10 Velocity. The average muzzle velocity shall not vary more than ± 20 meters per second (mps) from that noted on the contractor's top level drawing, listed on dwg. 12556500, when the cartridge is conditioned at 21° ± 3°C and fired in an M242 or 25mm Mann Barrel. In addition, the cartridge standard deviation shall not exceed 13 mps (see 4.5.5).

3.11 Waterproofness. There shall be no evidence of water entry into the propellant of any cartridge. Essential safety requirements (average peak chamber pressure plus 3.0 standard deviations shall not exceed 496 MPa and action time) shall be met after exposure. The average velocity (wet) of sample cartridges conditioned at 21° ± 3°C, shall not vary from the average velocity (dry) by more than 30.5 mps (see 4.5.5). Action time shall not exceed 6 ms.

3.12 Dispersion - Mann barrel and M242 gun. The standard deviation(s) of projectile impacts in both the horizontal and the vertical directions of the M910 Cartridge, conditioned at temperatures from -32° to 52°C, shall not be greater than 0.484 milliradians (mr) when fired single shot from the M242 Weapon and shall not be greater than 0.40 mr when fired from the Mann Barrel. The standard deviation shall not be greater than 0.65 mr when fired at a rate of 100 shots per minute (spm) from the M242 Weapon (see 4.5.7).

The above requirements shall be used to evaluate the First Article Inspection samples. For evaluation of Quality Conformance Inspection samples the standard deviation(s) of impacts in both the horizontal and vertical directions shall not be greater than 0.484 mr when conditioned and fired at -32°C to -22°C and +42°C to +52°C and shall not be greater than 0.40 mr when conditioned and fired at -22°C to +42°C from the Mann barrel.

3.13 Time of flight. The time for the projectile to reach a range of 2000 meters shall be between 1.1 to 2.1 seconds, when the cartridge is conditioned and fired at any temperature from -32°C to 52°C (see 4.5.9).

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3.14 Projectile trace. The projectile shall provide a visible trace in both daylight and night conditions at ranges from a minimum of 100 to at least 2000 meters from the muzzle (see 4.5.8).

3.15 Salt fog. The cartridge shall show no evidence of corrosion and shall be safe to operate in the M242 weapon system after exposure to the required salt fog environment for 48 hours. Function and casualty and metal parts security requirements shall be met after exposure (see 4.5.12).

3.16 Humidity. The cartridge shall show no evidence of corrosion and shall be safe to operate in the M242 weapon system. It shall meet function and casualty and metal parts security requirements after exposure to cycling temperatures and relative humidity levels (see 4.5.13).

3.17 Function, casualty and metal parts security. The cartridge shall function without casualty or hazards when conditioned and fired from the 25mm M242 Weapon at 200 spm. For lot acceptance testing, the cartridges shall be conditioned to ambient temperature. For First Article tests, one half the sample shall be conditioned to -54°C and the other half to +71°C. The cartridge shall exhibit metal parts integrity in the weapon feed mechanisms (chuting, feeding and chambering). Components designed for separation such as the sabot, protective cap and base are excluded (see 4.5.10).

3.18 Ruggedness. The cartridge shall withstand the rigors of rough handling and transportation without any adverse effect on safety and functioning at any temperature from -54°C to 71°C. The cartridge shall meet pressure, velocity, action time, trace, dispersion, function and casualty and metal parts security requirements after the transportation vibration phase. The cartridge shall meet the function and casualty, and metal parts security requirements after being exposed to each of the other environmental phases in sequential testing (see 4.5.11).

3.19 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. All parts and assemblies shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil, corrosion products and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any part or assembly, nor shall

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they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings and stampings shall be neat and legibly defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 and MIL-A-2550 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the requirements and test method paragraphs in the specification.

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting activity. The first article sample shall be manufactured using the same equipment, processes and procedures as will be used in production.

4.3.2 Inspections to be performed. The examinations and tests listed in Table I shall be performed on the first article sample in accordance with the test methods prescribed in 4.5. Unless otherwise specified, tests shall be conducted at 20° ± 10°C.

4.3.3 Rejection. If any of the subassemblies or assemblies fail to comply with the requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any component, subassembly or assembly to comply with any of the stated requirements. In the event of rejection, the Government reserves to right to require the contractor to take corrective action and submit a new first article quantity.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, 25mm, M910, TPDS-T	1 OF 4	12556500
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cartridge, 25MM, M910, TPDS-T Examination for Defects	As required	3.3	4.4.2 [1, 2]
	Projectile torque	32 [3] Acc-0 Rej-1	3.4	4.5.2
	Projectile extraction	32 [3] Acc-0 Rej-1	3.5	4.5.3
	Propellant contamination	32 [4] Acc-0 Rej-1	3.6	4.5.4
	Propellant weight	32 [4] Acc-0 Rej-1	3.7	Scale

NOTES:

[1] Examinations for special defects shall be performed on the entire lot. The occurrence of a special defect requires a failure analysis before proceeding with production.

[2] MIL-STD-651 shall apply in defining and evaluating visual defects, as applicable.

[3] Single sample size may be used and test combined provided no movement is experienced in the torque test, which shall be performed prior to extraction testing.

[4] Samples from torque and extraction tests shall be used.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 4	DRAWING NUMBER
	Cartridge, 25mm, M910, TPDS-T		12556500
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Chamber pressure [5]		3.8	4.5.5
	a. -54° ± 3°C	32 [5a]		
	b. 21° ± 3°C	32 [5b]		
	c. 71° ± 3°C	32 [5a]		
	d. -54° ± 3°C	32 [5a]		
	Velocity [5]		3.10	4.5.5
	21° ± 3°C	32		
	Action time [5]		3.9	4.5.6
	a. -54° ± 3°C	32 Acc-0 Rej-1		
	b. 21° ± 3°C	32 Acc-0 Rej-1		
	c. 71° ± 3°C	32 Acc-0 Rej-1		
	d. -54° ± 3°C	32 Acc-0 Rej-1		

NOTES:

[5] Velocity, Pressure and Action Time test data shall be gathered concurrently in combination testing. In addition, test d. shall be conducted with a reverse Lake City Roll (powder to bullet).

[5a] Criteria $\bar{x} + 2\sigma$ and no value greater than 496 MPa

[5b] Criteria $\bar{x} + 3\sigma < 454$ MPa

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 3 OF 4	DRAWING NUMBER
	Cartridge, 25mm, M910, TPDS-T		12556500
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA		REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Waterproofness [6] 21° ± 3°C	20	Acc-0 Rej-1	3.11	4.5.5
	Dispersion			3.12	4.5.7
	Mann Barrel 2000 meters [7]			3.12	4.5.7.1
	a. -32° ± 3°C	20			
	b. 21° ± 3°C	20			
	c. 52° ± 3°C	20			
	M242 Single Shot 2000 meters [8]			3.12	4.5.7.2
	a. -32° ± 3°C	20			
	b. 21° ± 3°C	20			
	c. 52° ± 3°C	20			
	M242 Burst (100 spm) 2000 meters [9]			3.12	4.5.7.3
	a. -32° ± 3°C	20			
	b. 21° ± 3°C	20			
	c. 52° ± 3°C	20			
	Time of flight [10]			3.13	4.5.9
	a. -32° ± 3°C	20	Acc-0 Rej-1		
	b. 21° ± 3°C	20	Acc-0 Rej-1		
	c. 52° ± 3°C	20	Acc-0 Rej-1		

NOTES:

[6] All twenty cartridges shall be torn down and examined for evidence of intrusion and effects of environment.
[7] Criteria: Standard deviation in X-direction .40
Standard deviation in Y-direction .40
[8] Criteria: Standard deviation in X-direction .484
Standard deviation in Y-direction .484
[9] Criteria: Standard deviation in X-direction .65
Standard deviation in Y-direction .65
[10] This data shall be gathered concurrently with Mann Barrel Dispersion data at each temperature.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70775A (AR)

PARAGRAPH	TITLE	SHEET 4 OF 4	DRAWING NUMBER
	Cartridge, 25mm, M910, TPDS-T		12556500
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Projectile trace [11]		3.14	4.5.8
	a. -32° ± 3°C	45 Acc-0 Rej-1		
	b. 21° ± 3°C	45 Acc-0 Rej-1		
	c. 52° ± 3°C	45 Acc-0 Rej-1		
	Salt fog [12]	20 Acc-0 Rej-1 10 teardown	3.15	4.5.12
	Function, casualty and metal parts security [13]		3.17	4.5.10 (Table IV)
	a. -54° ± 3°C	100 Table IV		
	b. 71° ± 3°C	100 Table IV		
	Humidity	110 Acc-0 Rej-1	3.16	4.5.13
	Ruggedness	480	3.18	4.5.11

NOTES:

[11] Blinds and short burn times in excess of the acceptance number are not permitted.
[12] Ten cartridges shall be torn down and examined for evidence of intrusion after completion of conditioning.
[13] The cartridge lot shall be rejected if any malfunction or firing casualty of Table IV occurs in number(s) equal to or greater than the applicable "Rej" number for the total sample (hot and cold combined). Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a sample of 400 cartridges (200 each at -54° ± 3°C and 71° ± 3°C) shall be selected. The lot shall be rejected if in the combined samples (first and second sample, hot and cold combined), malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The inspection lot shall comply with the lot formation provisions of MIL-A-48078. In addition, each cartridge lot offered for acceptance shall contain:

a. Primers from one lot interfix number from one manufacturer.

b. Metal parts from one lot interfix number from one manufacturer.

c. Boosters from one lot interfix number from one manufacturer.

d. Propellant from one lot from one manufacturer.

e. Tracer mixtures from one lot interfix number from one manufacturer.

f. Projectile, sabot and protective cap molding from one lot interfix number from one manufacturer.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection.

Lot Size	Inspection Levels III	IV
2 to 8	*	*
9 to 15	*	*
16 to 25	*	*
26 to 50	*	32
51 to 90	50	32
91 to 150	50	32
151 to 280	50	32
281 to 500	50	48
501 to 1200	75	73
1201 to 3200	116	73
3201 to 10000	116	86
10001 to 35000	135	108
35001 to 150000	170	123
150001 to 500000	200	156
500001 and over	244	189

NOTE: Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, 25mm, M910, TPDS-T	1 OF 2	12556500
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined.			
SPECIAL				
1.	Split or perforated case	100%	3.2	Visual
2.	Crimp missing (case to projectile)	100%	3.2	Visual
3.	Low propellant (25% below established charge weight)	100%	3.2	Visual
MAJOR	[2]			
101.	Overall length, maximum	Level III	3.2	SMTE
102.	Profile and alignment [1]	Level III	3.2	Dwg. 9391399
103.	Improper depth of primer seating	Level III	3.2	SMTE
104.	Cocked, loose or inverted primer	Level III	3.19	Visual
105.	Crimp improper or incomplete (case to projectile)	Level III	3.19	Visual
106.	Waterproofing missing around primer	Level III	3.19	Visual
107.	Improper head configuration	Level III	3.19	Visual

NOTES:
[1] A dead weight load of 60 pounds (270 Newtons) may be used to insert the cartridge into the gage.
[2] MIL-STD-651 shall apply in defining and evaluating visual defects, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.1	Cartridge, 25mm, M910, TPDS-T		12556500
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MINOR				
201.	Missing, damaged or cracked plastic parts	Level IV	3.19	Visual
202.	Corrosion	Level IV	3.19	Visual
203.	Evidence of corrosion	Level IV	3.19	Visual
204.	Marking incorrect, incomplete illegible or missing	Level IV	3.19	Visual
205.	Presence of foreign matter	Level IV	3.19	Visual
206.	Cartridge case failures (i.e., case draw, case dent, fold, bulge, wrinkle, or buckle)	Level IV	3.19	Visual

NOTES:

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4.4.3 Testing. The tests listed in Table III shall be performed on each cartridge lot conforming to 4.4 in accordance with the methods of inspection prescribed in 4.5. Unless otherwise indicated in the methods of inspection, tests shall be conducted with samples at 20° ± 10°C. Sample size and acceptance criteria for each test shall be as specified. Only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be thoroughly mixed before being divided into samples for the various tests. The combining of tests is permitted. The lot shall be rejected if in any firing test one or more projectiles remain in the gun bore. If a firing defect of Table IV occurs which would prevent the obtaining of a reliable result for the characteristics being tested, a replacement shot shall be fired. These firing defects or casualties found, whether from a Mann barrel or M242 weapon, shall be added to any found during function, casualty and metal parts security testing (see 3.17).

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TABLE III. Tests for quality conformance inspection.

Test	No. of Sample Units	Requirement Paragraph	Method Paragraph	Acceptance Criteria
Projectile torque	20	3.4	4.5.2	[1] 20 0,2
Projectile extraction	20	3.5	4.5.3	[2] 40 1,2
Propellant contamination	20	3.6	4.5.4	Acc-0/Rej-1
Chamber pressure				
-32° ± 3°C	20	3.8	4.5.5	[3]
21° ± 3°C	20	3.8	4.5.5	[3]
52° ± 3°C	20	3.8	4.5.5	[3]
Action time				
-32° ± 3°C	20	3.9	4.5.6	Acc-0/Rej-1
21° ± 3°C	20	3.9	4.5.6	Acc-0/Rej-1
52° ± 3°C	20	3.9	4.5.6	Acc-0/Rej-1
Velocity				
21° ± 3°C	20	3.10	4.5.5	[4]
Waterproofness				
21° ± 3°C	20	3.11	4.5.5	[5]
Dispersion [6]				
-32° ± 3°C	40	3.12	4.5.7	[7]
21° ± 3°C	40	3.12	4.5.7	[7]
52° ± 3°C	40	3.12	4.5.7	[7]
Projectile trace				
-32° ± 3°C	35	3.14	4.5.8	[8]
21° ± 3°C	35	3.14	4.5.8	[8]
52° ± 3°C	35	3.14	4.5.8	[8]
Function, casualty and metal parts				
Ambient [9]	200	3.17	4.5.10	Table IV

Refer to next page for "notes."

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TABLE III. Tests for quality conformance inspection. (NOTES)

[1] Failure of two or more of the sample cartridges to comply with the torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fail to comply with the applicable requirement.

[2] Failure of two or more sample cartridges to comply with the minimum extraction requirement shall be cause for rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fail to comply with the minimum requirement.

[3] If the sample fails to comply with the chamber pressure requirement, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the second sample fails to comply with the requirement.

[4] If the sample fails to comply with the velocity requirements, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the second sample fails to comply with the applicable requirements.

[5] All cartridges will be fired for comparison with velocity requirements. If the average velocity (wet) of the waterproofness sample fails to comply with the requirement, the lot shall be rejected. If any sample cartridge fails to meet the action time requirement or the pressure requirement, the lot shall be rejected.

[6] Fire from a Mann Barrel with muzzle brake at a range between 100 to 200 meters.

[7] If the dispersion of the sample cartridges exceeds the applicable requirement, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the dispersion of the second sample exceeds the applicable requirements.

[8] The lot shall be accepted if 0 failures occur in the first sample. If one unit fails to comply, a second sample of 35 cartridges shall be tested. The lot shall be rejected if 3 or more cartridges of the combined sample fail to comply with the minimum requirements. These provisions apply to blinds and short burn times.

[9] The cartridge lot shall be rejected if any malfunction or firing casualty of Table IV occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridges shall be selected. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number. The cartridge lot shall also be rejected for the conditions listed in 4.5.11.

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TABLE IV. Firing defects.[5]

Defect Description	Acc	Rej
Cartridge:		
Misfire	0	1
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1
In bore sabot separation	0	1
Primer:		
Primer leak	9	25
Primer perforation	1	6
Loose primer	0	1
Blown primer	0	1
Case:		
Longitudinal split [1]		
H or S	10	31
G or J	4	31
K, L or M	0	1
Circumferential rupture (partial) [1]		
S, J or K	4	11
G or L	2	7
Circumferential rupture (complete) [1]	0	1
Detached metal [2]	0	1
Projectile:		
Failure of sabot to separate [3]	0	4
Metal parts separation [4]	0	1
Trace fallout	1	2

Refer to next page for "notes."

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TABLE IV. Firing defects. (NOTES)

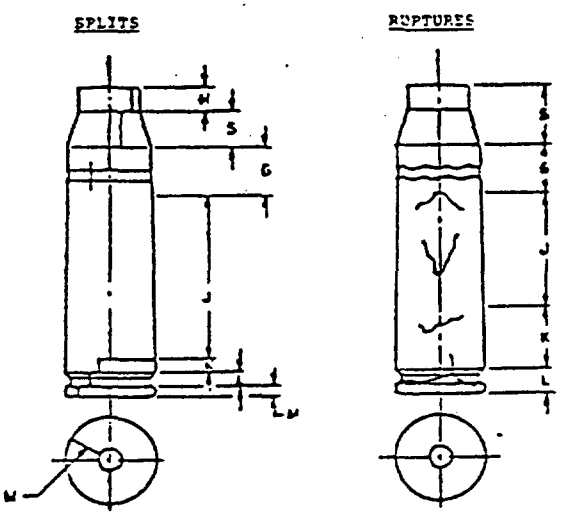
[1] See Figure 1 for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table IV for the areas involved shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classified as a complete circumferential rupture.

[2] Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears, shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

[3] Failure of the sabot to separate, as evidenced by penetration of the target screen, shall be classified as a defect.

[4] Break up of metal parts as evidenced by evaluation of holes in paper screen, shall be classified as a defect.

[5] Defect definitions are contained in Section 11 of AS12013566.



G - In Link Groove Section

H - In Mouth or Neck

J - In Main Body Section, No Closer than 1/4 inch to Extractor

K - In Lower Body Section From Forward Edge of Extractor Groove

L - In Extractor Groove

M - In Case Head

S - In Shoulder Section

G - In Link Groove Section

S - In Neck and Shoulder

J - In Upper 3/4 Body Section

K - In Lower 1/4 Body Section Above Extractor Groove

L - In Head and Extraction Groove

FIGURE 1. Classification of defects for case, cartridge, 25mm.

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4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.4.4.1 Packaging inspection. The sampling and inspection of the packaging, packing and marking for conformance with 5.1 and 5.2 shall be in accordance with MIL-P-71136.

4.4.4.2 Unlisted firing defects. Ammunition lots shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any testing indicates that the lot is unsuited for the purpose intended.

4.5 Methods of inspection.

4.5.1 Profile and alignment, max. Each cartridge assembly shall be inserted into the chamber gage which conforms to Drawing 9391399 to check for profile and alignment. The maximum force to insert the cartridge assembly into the gage shall be 60 pounds. Discrepant cartridges shall be removed from the lot.

4.5.2 Projectile torque. Each cartridge of the test sample shall be marked with a black ink marker. The mark, shall extend axially across the projectile assembly to the case. The torque specified shall be applied gradually. Movement of the projectile assembly with respect to the cartridge as evidenced by the misalignment of the mark shall be classified as a defect and cause for rejection of the lot.

4.5.3 Projectile extraction. The method of test shall be as specified in AS12013566, Section 12. Failure to satisfy requirements shall be cause for rejection of the lot.

4.5.4 Propellant contamination. The cartridge assemblies shall be examined visually for contamination as prescribed in Section 12 of AS12013566. Evidence of contamination in excess of that prescribed on the respective propellant drawing shall be cause for rejection of the lot.

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4.5.5 Velocity, pressure and waterproofness. The cartridges shall be tested in accordance with Section 5 of AS12013566. Reference cartridges shall be used for correction factors of velocity and pressure. Velocity and pressure data shall be gathered concurrently with action time data. Failure to satisfy the requirements shall be cause for rejection of the lot. The following additional provisions shall apply to the Waterproofness Test.

a. First article test (see Table I). The cartridges will be disassembled and the projectiles, propellant and cartridge case interior will be examined under ultraviolet light for evidence of leakage. Evidence of water entry into the propellant shall be cause for rejection of the lot. Leakage paths and other effects of water entry shall be noted. Photographs shall be taken, as required.

b. Quality conformance test (see Table III). Fire as specified in 4.5.5, using drilled cases, and compare the average muzzle velocity of the waterproofness tested (wet) cartridges with the average muzzle velocity obtained for the dry tested cartridges. The average muzzle velocity of the waterproofness cartridges shall not vary from the average velocity of the dry cartridges tested above by more than 30.5 mps. Pressure and action time requirements must be met. Failure to satisfy acceptance criteria shall be cause for rejection of the lot. (**NOTE:** Sodium fluorescinate shall not be used with this test.)

4.5.6 Action time. The method of test shall be as prescribed in Section 5 of AS12013566. Failure to satisfy requirements shall be cause for rejection of the lot.

4.5.7 Dispersion. The cartridges shall be tested in accordance with AS12013566, Section 7. Firing shall be done to determine dispersion characteristics from a firm fixed mount, unless otherwise specified. All cartridges shall be fired at an approximate 30-foot by 30-foot vertical target or other suitable target to insure that all cartridges fired shall impact the target. Any round that fails to hit the target shall be cause for rejection of the lot. The target shall be flat without sagging or wrinkles. Before testing, a sufficient number of spotting and warming cartridges shall be fired to condition the weapons and to ensure that the cartridges are likely to hit the target. Impact coordinates for each cartridge, rate of fire, weapon functioning and other pertinent information shall be recorded. The standard deviation of each cartridge group shall

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be recorded. The standard deviation of each cartridge group shall be calculated in both the horizontal and vertical directions. The computation shall be in accordance with Section 7 of AS12013566. Discrepancies from requirements shall be cause for rejection of the lot.

4.5.7.1 Mann barrel test. The sample cartridges shall be conditioned for a minimum of four hours at -32° ± 3°C, 21° ± 3°C and 52° ± 3°C and tested at 2,000 meters. This method of inspection applies to evaluation of first article inspection samples. Testing of quality conformance samples may be tested at 100 to 200 meters.

4.5.7.2 M242 single shot tests. The sample cartridges shall be conditioned for a minimum of four hours at -32° ± 3°C, 21° ± 3°C, and 52° ± 3°C and tested at 2,000 meters. All cartridges shall be fired in the single shot mode.

4.5.7.3 M242 burst fire tests. The sample cartridges shall be conditioned for a minimum of four hours at -32° ± 3°C, 21° ± 3°C, and 52° ± 3°C and tested at 2,000 meters. The cartridges shall be fired at a rate of 100 spm in five (5) round bursts for the listed number of cartridges at each temperature. The test shots shall be fired without adjustment to the weapon between bursts for each temperature group.

4.5.8 Projectile trace. The cartridges shall be tested in accordance with Section 8 of AS12013566. The cartridges shall be conditioned for a minimum of four hours at -32° ± 3°C, 21° ± 3°C and 52° ± 3°C and fired from the M242 Weapon in the single shot mode. Weapon elevation will ensure a minimum of a 4000 meter projectile free flight before ground impact. Time of trace and range of trace functioning shall be determined as measured from the muzzle of the weapon. Blinds, tracer fall-outs and short burn times in excess of the acceptance numbers are not permitted and shall be cause for rejection of the lot.

4.5.9 Time of flight. The cartridges shall be tested in accordance with TECOM ITOP 4-2-805 and MTP 4-2-827 or a method with an equivalent accuracy. Projectiles shall be tracked at ranges up to 2000 meters minimum. The cartridges shall be conditioned for a minimum of four hours at -32° ± 3°C, 21° ± 3°C and 52° ± 3°C and fired from a Mann Barrel. Failure to satisfy requirements shall be cause for rejection of the lot.

4.5.10 Function, casualty and metal parts security. The cartridges shall be tested in accordance with AS12013566, Section 11. The M242 Weapon used shall have a barrel with

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greater than 4,000 rounds fired through it. A vertical witness screen shall be positioned approximately 60m to 200m normal to the line of fire such that metal part failure data may be obtained. The cartridges shall be conditioned for a minimum of four hours at -54° ± 3°C, 21° ± 3°C and 71° ± 3°C. When ambient temperature conditions are 20° ± 10°C, testing for the 21° ± 3°C group can be at range ambient conditions. All firings shall be at 200 ± 50 spm in bursts of 21 to 28 rounds. The weapon shall be cooled to ambient temperature after each temperature and range group test is completed (100 maximum cartridges fired after which cooling to ambient is required.) The witness screen shall be examined during the cooling period for fragment imprints and metal parts security failures. Evidence of yaw or other erratic flight characteristics shall be recorded and evaluated for cause. Witness screen replacement shall be provided as needed. Firing defects specified in Table IV shall also be determined during the cooling period. Any weapon stoppages obtained shall be recorded and evaluated for cause. Firing defects in excess of the quantities listed in Table IV shall be cause for rejection of the lot. The lot shall be rejected if 20% of the rounds fired exhibit more than 15 degrees yaw.

4.5.11 Ruggedness. A sequential ruggedness test will be performed on 480 sample cartridges. This test will be conducted in accordance with the outline in Table V and will consist of the following phases in order of occurrence: basic transportation (secured cargo), 2.1 meter packaged drop, loose cargo and 1.5 meter unpacked drop. A sample of cartridges will be withdrawn after each test phase for test firing from an M242 Weapon or a Mann Barrel. The test firings will be conducted at 21°C ± 10°C with the M242 fired in the 200 spm mode. Metal parts security data will be recorded for evaluation.

4.5.11.1 Basic transportation (secured cargo). This test will be conducted in accordance with MIL-STD-810, Method 514, Category 1. Before the test, each cartridge will be visually inspected for damage from previous transportation and handling, as well as, any obvious irregularities in packing. Two hundred and forty cartridges shall be conditioned at -54° ± 3°C and a like amount of cartridges shall be conditioned 71° ± 3°C for 12 hours prior to being vibrated at their respective temperature environment. The cartridges will be packed in ammunition containers and will be tested as packed for shipment.

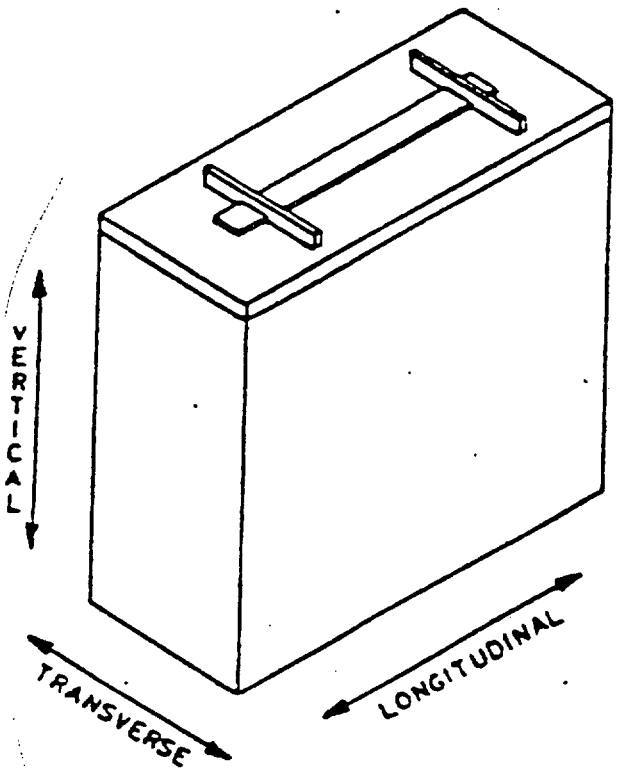
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At the conclusion of all vibration tests, the external surfaces of all cartridges and their packing materials will be visually inspected for evidence of deformation, damage or unusual defects that would render them unsafe to handle, transport, or fire (see 4.5.11.5). Any cracking or splitting of the cartridge plastic protective cap shall be cause for rejection of the lot. Ninety cartridges from each temperature shall then be randomly selected and fired per the requirements of 3.8, 3.9, 3.10, 3.12, 3.13 and 3.17.

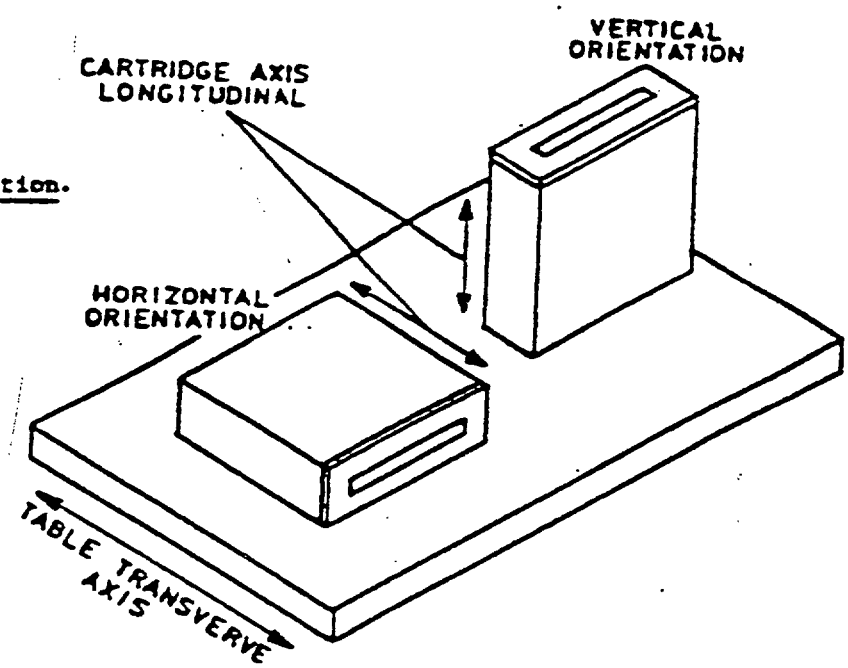
4.5.11.1.1 Transportation vibration phase (secured cargo). Cartridges shall be subjected to Wheeled Vehicle schedules, Two Wheeled Vehicle schedules, and Tracked Vehicle schedules in accordance with MIL-STD-810, Method 514, Category 1.

4.5.11.2 2.1 Meter packaged drop phase. This phase shall be conducted in accordance with TECOM TOP 4-2-602 and follows the basic transportation phase for secured cargo. The packaging shall be dropped on the three axes defined in Figure 2. One hundred and fifty cartridges, packaged, shall be conditioned at -54°C ± 3°C and one hundred and fifty cartridges, packaged, shall be conditioned at 71°C ± 3°C for 12 hours prior to testing. At the conclusion of this test, the external surfaces of all cartridges and their packing materials will be visually inspected for evidence of deformation, damage or unusual effects that would render them unsafe to handle, transport, or fire (see 4.5.11.5). Any cracking or splitting of the cartridge plastic protective cap in excess of two percent of the test sample shall be cause for rejection of the lot. Forty-five (one box and one 15-round belt) cartridges shall then be randomly selected from each temperature environment and fired per the requirements of 3.17.

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Transportation vibration axes definition.



Loose cargo vibration axes definition.

FIGURE 2. Directional axes.

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4.5.11.3 Loose cargo phase. After completion of the 2.1 meter packaged drop phase and removal of the sample for firing, the remaining cartridges shall be inspected, repackaged and then subjected to the loose cargo phase. Testing will be in accordance with MIL-STD-810, Method 514, Category 3. Ninety repackaged and the one loose belt of fifteen cartridges shall be conditioned at -54°C ± 3°C and a similar quantity and configuration of cartridges shall be conditioned at 71°C ± 3°C for 12 hours prior to being subjected to this test.

At the conclusion of the loose cargo phase, the external surfaces of all cartridges and their packing materials will be visually inspected for evidence of deformation, damage or unusual defects that would render them unsafe to handle, transport or fire (see 4.5.11.5). Any additional cracking or splitting of the cartridge plastic protective cap or propagation of any previous cracks or splits shall be cause for the rejection of the lot. Forty-five cartridges from each temperature environment shall then be randomly selected and fired per the requirements of 3.17.

4.5.11.4 1.5 Meter unpackaged drop phase. This is the final phase of the sequential ruggedness testing. Testing will be performed in accordance with TECOM TOP 4-2-602. After inspecting the cartridges previously subjected to the loose cargo phase, the test samples shall be delinked and quantities shall be conditioned at -54°C ± 3°C and 71°C ± 3°C, respectively, for 12 hours prior to conducting the test. After the drop test, the cartridges shall be inspected for damage. If judged to be safe to handle and fire, the cartridges shall be relinked and fired per the requirements of 3.17.

4.5.11.5 Sabot damage. Sabot damage that would render M910 cartridge unsafe to handle, transport or fire during ruggedness testing is defined as, but not limited to the following:

a. Missing quandrant of the sabot

b. Missing segment of the rotating band

c. Missing segment of the bourrelet

d. For cartridge without positive subprojectile retention mechanism, circumferential cracks exceeding 270 degrees around the rotating band area.

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TABLE V. Ruggedness (rough handling) outline.

PART A - Transportation Vibration Phase (Secured Cargo) (4.5.11.1)

8 Containers (240 Cartridges) - 54°C
Visual inspection for damage
Secured cargo vibration
Photos and visual inspection
Remove 3 containers, mark each with -54 A and fire per Table VI

8 Containers (240 Cartridges) 71°C
Visual inspection for damage
Secured cargo vibration
Photos and visual inspection
Remove 3 containers, mark each with 71 A and fire per Table VI

PART B - 2.1 Meter (7 Foot Packaged Drop Phase) (4.5.11.2)

5 Containers (150 Cartridges) -54°C
Std Pkg(s) dropped once in each of 3 orientations; bottom, side and latch down. Inspect.
Photos and visual inspection
Remove 1 container and 15 Cartridges; mark each with -54 B and fire per Table VI

5 Containers (150 Cartridges) 71°C
Std Pkg(s) dropped once in each of 3 orientations; bottom, side and latch down. Inspect.
Photos and visual inspection
Remove 1 container and 15 Cartridges; mark each with 71 B and fire per Table VI

PART C - Loose Cargo Phase (4.5.11.3)

3 Cont. plus 1-15 Rd Belt (105 Cart.) -54°C
90 Cart., 1 Belt, Vert. and Horiz. Vibration
Photos and visual inspection
Remove 1 Container and mark with -54 C1 and fire per Table VI
Remove 15-round belt and mark each round with -54 C2 and fire per Table VI

3 Cont. 1-15 Rd Belt (105 Cart.) 71°C
90 Cart., 1 Belt, Vert. and Horiz. Vibration
Photos and visual inspection
Remove 1 Container and mark with 71 C1 and and fire per Table VI
Remove 15-round belt and mark each round with 71 C2 and fire per Table VI

PART D - 1.5 Meters (5 Foot) Unpackaged Drop Test (4.5.11.4)

60 Cartridges (Rds) -54°C

10 Rds Drop ea Rd. in 5 Orientations	Drop 10 Rds Nose Down	Drop 10 Rds Base Down	Drop 10 Rds Horiz.	Drop 10 Rds 45° Base Down	Drop 10 Rd 45° Nose Down
Mark D1	Mark D2	Mark D3	Mark D4	Mark D5	Mark D6

Photos, visual insp req'd; fire per Table VI

60 Cartridges (Rds) 71°C

10 Rds Drop ea Rd. in 5 Orientations	Drop 10 Rds Nose Down	Drop 10 Rds Base Down	Drop 10 Rds Horiz.	Drop 10 Rds 45° Base Down	Drop 10 Rds 45° Base Down
Mark D7	Mark D8	Mark D9	Mark D10	Mark D11	Mark D12

Photos, visual insp req'd; fire per Table VI

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TABLE VI. Ruggedness firing schedule.

Marking Ident.	Qty	Weapon	Notes	Test Data [1]
-54 A	40	Mann Barrel	2	Vm, P, Ta, D, Trace
71 A	40	Mann Barrel	2	Vm, P, Ta, D, Trace
-54 A	50	M242 200 spm	2	MPS
71 A	50	M242 200 spm	2	MPS
-54 B	45	M242 200 spm	3	MPS
71 B	45	M242 200 spm	3	MPS
-54 C1	30	M242 200 spm	4	MPS
71 C1	30	M242 200 spm	4	MPS
-54 C2	15	M242 200 spm	4	MPS
71 C2	15	M242 200 spm	4	MPS
-54 D1	10	M242 200 spm	5	MPS
-54 D2	10	M242 200 spm	5	MPS
-54 D3	10	M242 200 spm	5	MPS
-54 D4	10	M242 200 spm	5	MPS
-54 D5	10	M242 200 spm	5	MPS
-54 D6	10	M242 200 spm	5	MPS
71 D7	10	M242 200 spm	5	MPS
71 D8	10	M242 200 spm	5	MPS
71 D9	10	M242 200 spm	5	MPS
71 D10	10	M242 200 spm	5	MPS
71 D11	10	M242 200 spm	5	MPS
71 D12	10	M242 200 spm	5	MPS

LEGEND

Vm	=	Muzzle Velocity
P	=	Chamber Pressure
Ta	=	Action Time
D	=	Dispersion
Trace	=	Trace Ignition, Burntime and Duration
MPS	=	Function, Casualty and Metal Parts Security

NOTES

[1] All rounds conditioned four hours minimum at 21 ° ± 3°C prior to firing.

[2] Three containers (6 belts) for -54 A and three containers (6 belts) for 71 A. Fire from Mann Barrel in 20-round groups at 2000 meters and from M242 in 25-round belts at approximately 60 to 200 meters, as specified in 4.5.11.1.

[3] Fire three 15-round belts for -54 B and three 15-round belts for 71 B from M242 Weapon at approximately 60 to 200 meters, as specified in 4.5.11.2.

[4] One container (2 belts) and a 15-round belt for -54 C and one container (2 belts) and a 15-round belt for 71 C. Fire in 15-round groups from M242 at approximately 60 to 200 meters, as specified in 4.5.11.3.

[5] Sixty cartridges for -54 D and sixty cartridges for 71 D. Fire in 10-round groups from M242 at approximately 100 to 200 meters, as specified in 4.5.11.4.

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4.5.12 Salt fog test. The cartridges in the M28 link belt shall be tested in accordance with MIL-STD-331, Test C-3, for 48 hours. After exposure to the salt fog environment, the cartridges shall be photographed and examined for evidence of contamination, damage, corrosion, and salt infiltration. Cartridges, as required, shall be torn down, photographed, and examined for evidence of intrusion and effects of the environment. The remaining cartridges shall then be ballistically tested using the M242 Weapon, fired single shot to determine whether the cartridge feeds and functions safely after exposure to the salt fog environment. The cartridge shall satisfy velocity and function, casualty and metal parts security requirements.

4.5.13 Temperature and humidity. One hundred and ten (110) cartridges shall be subjected to a ten-day temperature humidity storage test cycle as follows:

No. of Hours	Temperature °F (°C)	Relative Humidity (%)
2	Increased to +105 (41.5)	Increase to 85-90
16	Maintain at +105 ± 3 (41.5 ± 1.7)	Maintain 85-90
2	Decrease to 70 ± 3 (21 ± 1.7)	Increase to 97 + 3
4	Maintain 70 ± 3 (21 ± 1.7)	Maintain at 97 + 3

Repeat cycle ten times. During the four-hour period of exposure to 70° ± 3°F (21° ± 1.7°C) on the first, third, fifth, eighth and tenth days, remove twenty rounds and test from an M242 gun for compliance with function and casualty and metal parts security requirements. Cartridges removed on the first, third, fifth and eighth days should be marked for later firing with the cartridges removed on the tenth day. The ten remaining cartridges shall be torn down and examined for evidence of intrusion and effects of environment. Discrepancies from these requirements shall be cause for rejection of the lot.

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5. PACKAGING

5.1 Packing level A. The cartridge shall be linked into a belt and packed in accordance with instructions from the procuring agency. Linked ammunition shall be packed in accordance with Drawing 12929427 and MIL-P-71136 or 12013671 unless provided with other instruction by the procuring agency.

5.1.1 Packing level B. Same as 5.1.

5.1.2 Packing level C. Same as 5.1.

5.2 Marking. Marking on the packed container shall be in accordance with the Drawing 12929427 or 12013671 as appropriate.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridge is intended for target practice use in 25mm automatic weapons.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.5 Submission of alternate quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality assurance conformance provisions shall be submitted by the contractor to the contracting activity.

6.6 Subject term (key word) listing.

Nose Cap
Small Arms
Training

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project No. 1305-AE66)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-70775	2. DOCUMENT DATE (YYMMDD) 940215

3. DOCUMENT TITLE
CARTRIDGE, 25MM, TARGET PRACTICE DISCARDING SABOT WITH TRACER (TDPDS-T), M910

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. ARMY ARDEC STANDARDIZATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial: 201-724-6675 (2) AUTOVON: DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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